

LARRY "TRE" HUMPHRIES
ENGINEER & SYSTEMS BUILDER

[MENU](#)

ENGINEER & SYSTEMS BUILDER

I BUILD THE SYSTEMS BETWEEN THE MACHINE AND THE BUSINESS.

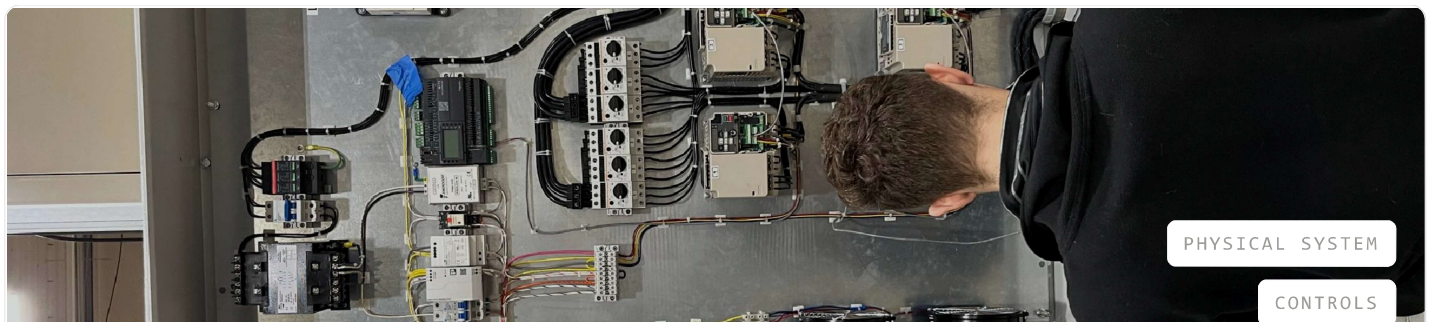
My background spans industrial equipment, controls, embedded systems, manufacturing, and internal software.

I'm usually most useful when a problem crosses those boundaries and no single person clearly owns the whole system.

Mechanical engineering background. Field-tested controls experience. Software built around real operations.

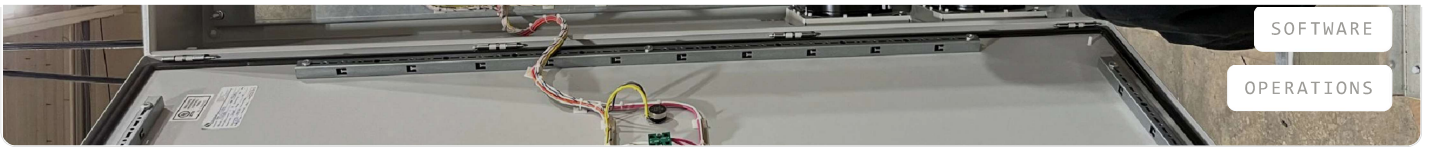
TELL ME ABOUT A PROBLEM

SEE WHAT I'VE BUILT



PHYSICAL SYSTEM

CONTROLS



WHOLE-SYSTEM VIEW

I WORK ACROSS THE WHOLE SYSTEM.

Most operational problems are not contained inside one discipline.

The equipment behaves one way. The controls describe another. The software captures only part of the picture. The people doing the work create their own process around the gaps.

My work is usually connecting those layers.

01

PHYSICAL SYSTEM

Equipment, motors, sensors, mechanisms, and field conditions.

02

CONTROLS

PLCs, HMIs, VFDs, Codesys, BACnet, Modbus, and motion systems.

03

SOFTWARE

Python, React, Flask, APIs, databases, dashboards, and internal tools.

04

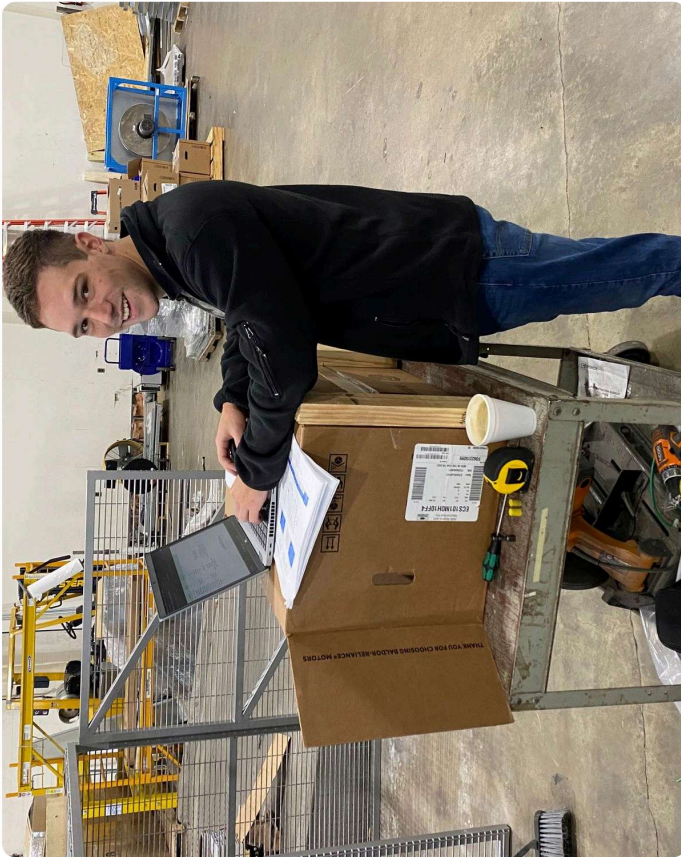
OPERATIONS

Quoting, configuration, testing, production, commissioning, and handoffs.

CAREER TIMELINE

I'VE SPENT MY CAREER WORKING AT THE SEAMS.

The industries and job titles changed, but the pattern stayed consistent: I kept getting pulled toward the problems between equipment, controls, software, and the people responsible for keeping everything moving.



SOMETIMES I'M LUCKY ENOUGH TO GET A BOX-DESK

21 - 22

2021-2022

CONTROLS ENGINEER

Innerspec Technologies



MILL-ROLL ROBOTIC INSPECTION PROJECT

Delivered industrial automation systems using Allen-Bradley, Siemens, and Beckhoff PLCs, motion systems, and field-integrated ultrasonic inspection equipment.

22-24

2022-2024

CONTROLS ENGINEER

Mainstream Fluid & Air

Led controls work for custom commercial HVAC fan-array systems using PLCs, HMIs, VFDs, BACnet, and Modbus, including startup, commissioning, quality, and field troubleshooting.

24-25

2024-2025

AUTOMATION ENGINEER

Mainstream Fluid & Air

Built internal quoting, fan-selection, dashboard, database, and process-automation tools using React, Python, Flask, PostgreSQL, REST APIs, authentication, and engineering logic.

25-26

2025-2026

SENIOR ENGINEERING & TECHNOLOGY DEVELOPER

GAF Roads / Standard Industries

Worked across Codesys controls, mechanical behavior, field troubleshooting, edge data, OTA readiness, cloud integration paths, and operator workflows on an early electric sprayer platform.

Across each role, I became increasingly responsible for the system around the system: the tools, information, configuration, testing, and operating processes that made the core product usable.

FEATURED WORK

SYSTEMS I'VE BUILT, IMPROVED AND STABILIZED.

The work varies in form: a control system, an internal platform, a commissioning workflow, or a business tool, but it usually begins with the same question:

What is making the real work harder than it needs to be?

PROJECT 1

TURNING PRODUCT KNOWLEDGE INTO A USABLE INTERNAL SYSTEM

FAN ARRAY SUBMITTAL



Project Name: Project Name

To: Client Name

Tag: Unit #

Array Qty: 1

Reference No.: 123123

Prepared By: Sales Engineer

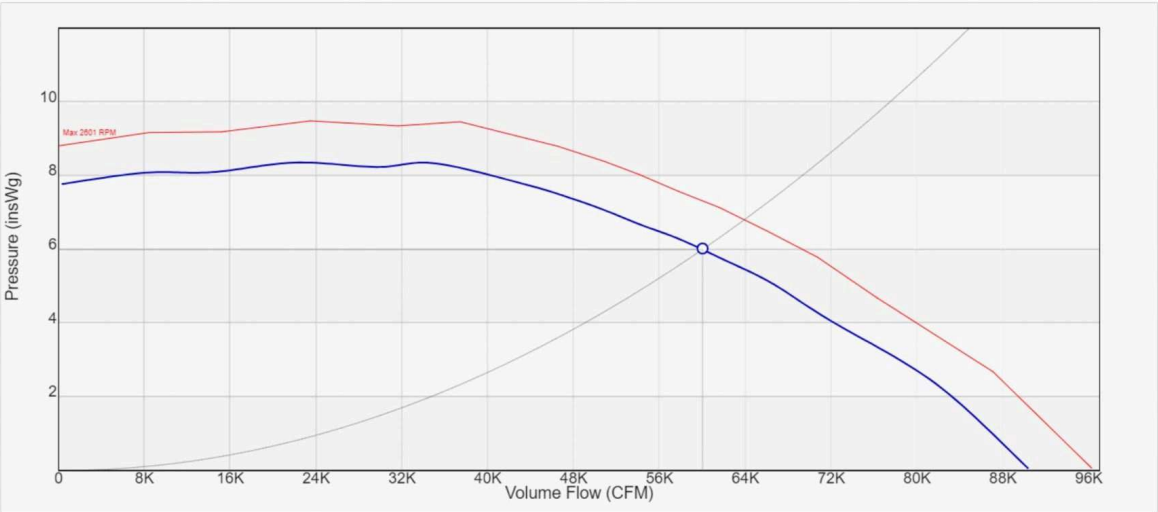
Date: 05/04/2023

PERFORMANCE

Airflow (total)	60,000 CFM
Qty of Fans/Array:	9
Total static pressure:	6 insWg
Op. point power (each):	8.26 HP (6.16 KW)
Nominal power (each):	10 HP (7.46 KW)
Operating RPM:	2441
Max RPM:	2601
FEI:	1.29

ELECTRICAL

Electrical supply:	3~ 460V 60Hz
Single fan op. Amp:	8.04
Array Op. Amp:	72.33
Single fan FLA:	12.5
Array FLA:	112.5
Array MCA:	115.63
Array MOCP:	125



Mainstream Fans are rated in accordance with AMCA Sound and Air Performance without accessories. Performance is shown for free Inlet/Free Outlet Installations.

Type:	63 Hz	125 Hz	250 Hz	500 Hz	1k Hz	2k Hz	4k Hz	8k Hz	A-Weighted
Discharge:	110.6	104.6	102.9	95.8	93.6	89.6	88.3	87.2	97.1
Inlet:	106.9	101.6	99.6	92.4	89.7	86.6	85.2	84.2	93.8



A technically complex product-selection and quoting workflow depended on spreadsheets, product rules, pricing logic, engineering knowledge, and handoffs.

BUILT

- React frontend
- Python/Flask backend
- PostgreSQL
- Authentication and roles
- Engineering calculations
- Pricing logic
- Product-selection workflow
- Windows-only vendor DLL integration

WHY IT MATTERED

It converted engineering knowledge into a clearer, maintainable internal system with better handoffs and less dependence on individual memory.

PROJECT 2

STABILIZING AN EARLY MACHINE IN REAL FIELD CONDITIONS



SPRAYER PLATFORM CONCEPT



FIELD-CONDITION REFERENCE

PROBLEM

An early electric sprayer platform needed stronger reliability and production readiness.

WORKED ACROSS

- Codesys controls
- Mechanical behavior
- Field failure investigation
- Edge data
- OTA readiness
- Cloud integration
- Operator workflows
- Cross-functional technical communication

WHY IT MATTERED

The issue was the whole operating system around the machine, not just one controller.

PROJECT 3

CONNECTING EQUIPMENT, CONTROLS AND THE BUILDING AROUND THEM



INSTALLED HVAC EQUIPMENT CONTEXT

PROBLEM

Configurable commercial HVAC fan-array systems needed practical controls, integration, commissioning, and field support.

BUILT AND SUPPORTED

- PLC/HMI logic
- VFD programming
- BACnet
- Modbus
- Product testing
- Startup
- Commissioning
- Quality support
- Field troubleshooting

WHY IT MATTERED

A controls design only works when it survives the drive, motor, equipment, building system, installer, and actual operating conditions.

ADDITIONAL BUILDS

ADDITIONAL BUILDS

PROJECT QUEUE DASHBOARD

Internal visibility for tracking work moving across teams and priorities.

SOFTWARE

OPERATIONS

DRIVE-PROGRAMMING WORKFLOW

Standardized workflow for configuring drives more reliably and repeatably.



DRIVE-PROGRAMMING HARDWARE REFERENCE

CONTROLS

WORKFLOW

MOTOR AND PRODUCT DATABASE

Structured product information for engineering, selection, and reference work.



MOTOR REFERENCE IMAGE

DATABASE ENGINEERING

MANUFACTURING AND ENGINEERING UTILITIES

Small internal tools that reduce repeated manual work across production and engineering.

MANUFACTURING UTILITIES

HOMEEMS DIGITAL AND LEAD-GENERATION SYSTEM

Digital workflow supporting lead intake and follow-up around a small business operating system.

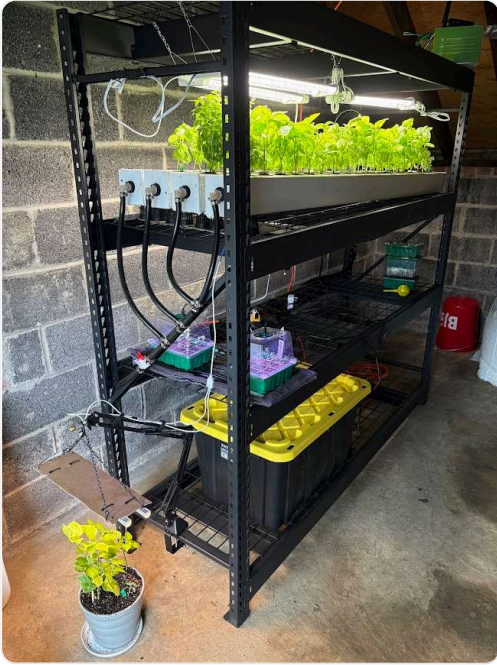


HOMEEMS IDENTITY

SOFTWARE BUSINESS

OPERATIONAL LOGGING AND FORECASTING SYSTEM

Internal logging and planning support for tracking demand, activity, and expected work.



VISOIN SUPPORTED GARAGE HORTICULTURE RACK

LOGGING

FORECASTING

PROBLEMS INVITATION

BRING ME THE PROBLEMS THAT FALL BETWEEN ROLES.

The best fit is usually not a neatly isolated software project or a clearly defined controls task.

It is a real operational problem touching several systems at once.

A product is difficult to configure, test, or commission

Important knowledge lives in a few people's heads

Equipment, controls, and software do not communicate cleanly

Production or engineering lacks visibility

A repeated task needs to become a tool

A problem crosses departments and nobody owns the whole thing

A small business has outgrown the systems it began with

WORKING APPROACH

START WITH THE REAL WORK.

Before deciding that the answer is a dashboard, control change, or new piece of software, I want to understand what the equipment does, how the process works today, and where the information or decisions stop lining up.

01

SEE THE WORK IN CONTEXT

Talk to the people doing it, inspect the equipment, and understand the current workflow.

02

FIND THE ACTUAL BOTTLENECK

Separate the visible frustration from the underlying technical or operational constraint.

03

BUILD THE SMALLEST USEFUL SYSTEM

Create enough structure to improve the work without overbuilding around assumptions.

04

TEST IT UNDER REAL CONDITIONS

Use the system with the equipment, data, and people it was designed to support.

05

IMPROVE AND MAKE IT MAINTAINABLE

Document the important logic, respond to real use, and reduce unnecessary dependence on the builder.

CONTACT

YOU DO NOT NEED A POLISHED BRIEF.

Send me the messy version.

Tell me what the equipment or team is trying to do, how the process works today, where it becomes difficult, and what you wish were easier.

I can usually tell fairly quickly whether my background is useful to the problem.

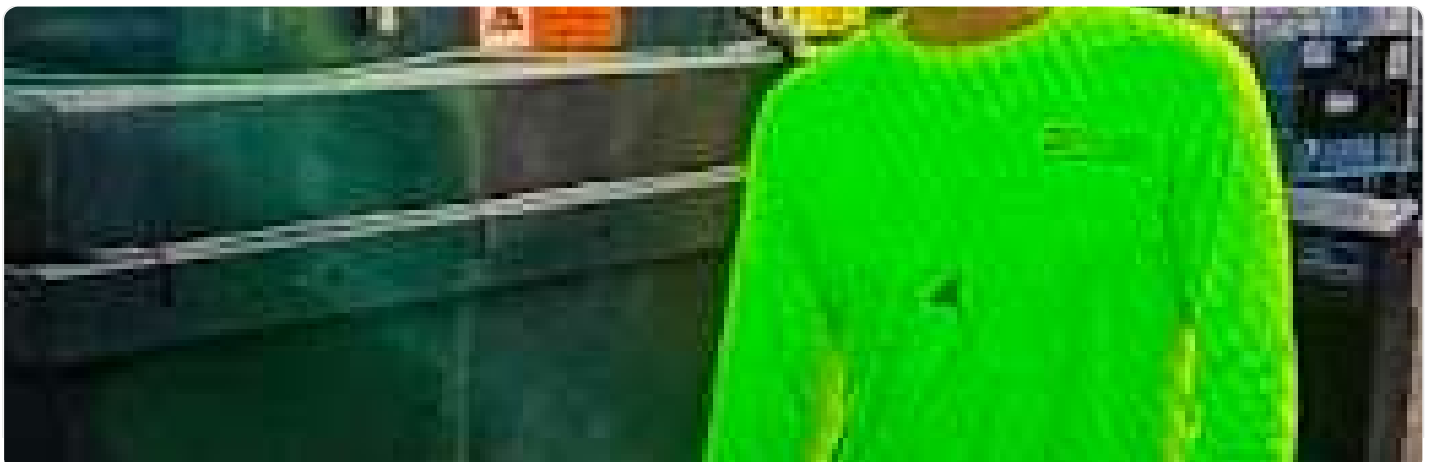
LARRY "TRE" HUMPHRIES

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NAME

Example text

COMPANY OR TEAM OPTIONAL

Example text

EMAIL

Example text

TELL ME ABOUT THE PROBLEM

Example text

Helpful context: how you handle it today, what makes it difficult, and what a better result would look like.

SUBMIT

LARRY "TRE" HUMPHRIES

Engineer & Systems Builder

Built around real systems.

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